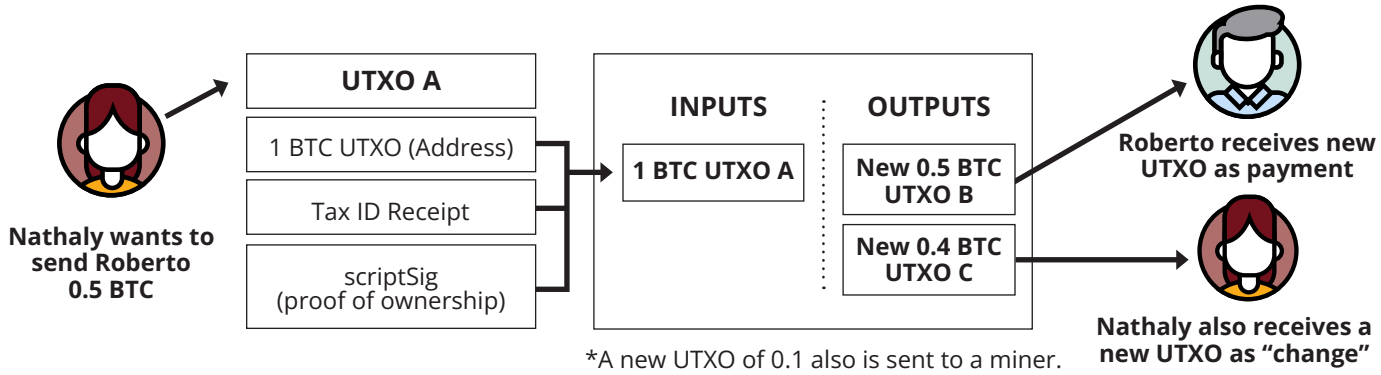
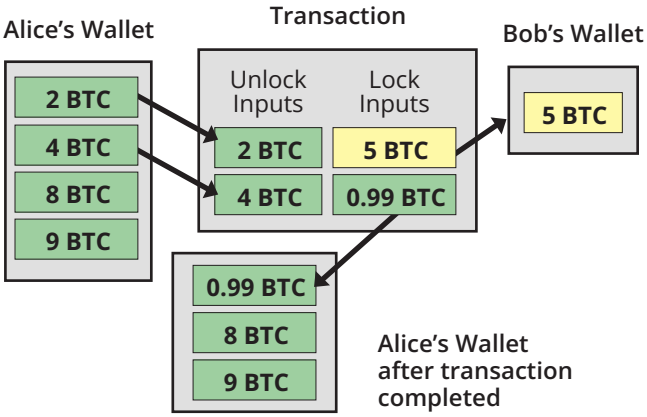


How UTXOs Work in Bitcoin Transactions

When a transaction is made, the amount of ***bitcoin*** that is sent is divided into multiple outputs, each of which is associated with a specific address.

- When sending ***bitcoin*** to someone, you will use one or more Unspent Transaction Outputs (UTXOs) as the source of the funds. These UTXOs will be combined, if necessary, to create a new output that belongs to the recipient of the transaction. This new output, or UTXO, will then become the recipient's property and can be used as the source of funds in a future transaction. This chain of UTXOs creates a transparent and traceable history of all ***bitcoin*** transactions on the *blockchain*, starting from the very first block.



- For example, if someone wants to send 2 ***bitcoin*** but has a UTXO worth 5 ***bitcoin***, the difference of 3 ***bitcoin*** is sent back to the sender as "change." This change is a new UTXO for the sender and can be spent in a future transaction.

- In the example, Alice sends Bob 5 BTC while retaining some for herself. She combines 6 ***bitcoin*** from her four UTXOs, which totals 23 ***bitcoin***, and sends 5 to Bob and .99 back to herself, with a .01 fee for processing. The transaction is then added to the *blockchain*, updating all nodes with a copy of the updated UTXO ledger. If Alice then tries to send 23 BTC to Ximena in a separate transaction, the nodes will reject it as some of the output has already been spent.

If someone were to attempt to use a spent output in their transaction, it would likely be rejected by the network nodes. This is because these nodes maintain a copy of the same database set and can easily reach a consensus by checking the balance of each address before validating any new transaction. This ensures the integrity and validity of the transactions on the network.